Example : Every a meromorphic function on the Riemann sphere $\mathbb{C}_\infty$ is a rational function

i.e. $\mathcal{M}(\mathbb{C}_\infty) = \mathbb{C}(z)$ $(:= \mathrm{Frac}(\mathbb{C}[z]))$

Proof. Clear that $\supset$ holds (partial fractions !)

For $\subset$, given $f \in \mathcal{M}(\mathbb{C}_\infty)$,

compactness of $\mathbb{C}_\infty$ $\Rightarrow$ the set P of poles of f is finite

(If P is infinite then it must have a limit point in compact $\mathbb{C}_\infty$, so identity theorem $\Rightarrow$ $1/f$ vanishes on $\mathbb{C}_\infty$, contradiction)

Let $a_1, \ldots, a_n \in \mathbb{C}$ (avoid ∞) be the poles of f in $\mathbb{C}$

For each index ν, let

$$h_\nu(z) = \sum_{j=-k_\nu}^{-1} c_{\nu j}(z - a_\nu)^j$$ (singular part) be the principal part of f at the pole $a_\nu \neq \infty$

At ∞, similarly let $w = 1/z$

$$g(w) = \sum_{j=-k_\infty}^{-1} c_{\infty, j}\, w^j$$ be the principal part of f at ∞

i.e. $$f(z) = \underbrace{\sum_{j=1}^{k} c_{\infty,-j}\, z^j}_{g(1/z)} + \text{power series in } 1/z$$ (polynomial in z, no const term)

Then the function $f(z) - \left(h_1(z) + \cdots + h_n(z) + g(1/z) \right)$ is meromorphic on $\mathbb{C}_\infty$ with no pole

$\Rightarrow$ is holomorphic on $\mathbb{C}_\infty$ $\Rightarrow$ is constant $c \in \mathbb{C}$ by Liouville's Theorem

$\Rightarrow$ $f(z) = c + \left(h_1(z) + \cdots + h_n(z) + g(1/z) \right) \in \mathbb{C}(z)$ is a rational function #

Additive Cousin Problem :

- Let G be a Riemann surface (eg. $G \subset \mathbb{C}$ an open set of $\mathbb{C}$ or $G = \mathbb{C}_\infty$ the Riemann sphere)
- Let $P \subset G$ be a closed subset with no limit point in G (pole set)
- For each $p \in P$, let $f_p \in \mathbb{C}[z]$ be a polynomial with no constant term
- **Does there exist** a meromorphic function f on G such that :
  - f is **holomorphic** on $G - P$
  - for each $p \in P$,
$$\begin{cases} f(z) - f_p\left(\dfrac{1}{z-p}\right) & \text{is holomorphic on a neighborhood of } p \quad \text{if } p \neq \infty \\ f(z) - f_\infty(z) & \text{is holomorphic on a neighborhood of } \infty \quad \text{if } \infty \in P \end{cases}$$
i.e.
$$\begin{cases} f_p\left(\dfrac{1}{z-p}\right) & \text{is the principal part of } f \text{ at } p \quad \text{if } p \neq \infty \\ f_\infty(z) & \text{is the principal part of } f \text{ at } \infty \quad \text{if } \infty \in P \end{cases}$$
?

- The corresponding uniqueness question is easy :

  If f is one such meromorphic function on G,
  then any such meromorphic function g on G is of the form
  $$g = f + (\text{holomorphic function on } G)$$

Compare :

Consider the following problem: Let G be an open subset of $\mathbb{C}$ and let $\{a_k\}$ be a sequence of distinct points in G such that $\{a_k\}$ has no limit point in G. For each integer $k \geq 1$ consider the rational function

3.1
$$S_k(z) = \sum_{j=1}^{m_k} \frac{A_{jk}}{(z-a_k)^j},$$

where m_k is some positive integer and $A_{1k}, \ldots, A_{m_k k}$ are arbitrary complex coefficients. Is there a meromorphic function f on G whose poles are exactly the points $\{a_k\}$ and such that the singular part of f at $z = a_k$ is $S_k(z)$?

Note :

- Since G is second countable, the set P is at most countable.
- If P is finite, then $f(z) := f_\infty(z) + \sum_{p \in P - \{\infty\}} f_p\left(\dfrac{1}{z-p}\right)$ is a solution
- More generally, for any finite subset $P_0 \subset P$,
  if there exists a solution f for $P - P_0$,
  then there exists a solution for P, namely $f(z) + \sum_{p \in P_0} f_p\left(\dfrac{1}{z-p}\right)$
  (interpret $\dfrac{1}{z-\infty}$ as z)

3.2 Mittag-Leffler's Theorem.

Let $G \subset \mathbb{C}_\infty$ be an open set of the Riemann sphere

Let $P \subset G$ be a closed subset with no limit point in G (pole set)

For each $p \in P$, let $f_p \in \mathbb{C}[z]$ be a polynomial with no constant term

Then there exists a meromorphic function f on G such that :

- f is holomorphic on $G - P$
- for each $p \in P$,
  - $f(z) - f_p\left(\frac{1}{z-p}\right)$ is holomorphic on a neighborhood of p if $p \neq \infty$
  - $f(z) - f_\infty(z)$ is holomorphic on a neighborhood of ∞ if $\infty \in P$

ie. on any open subset of $\mathbb{C}_\infty$, every **Additive** Cousin **Problem** has a solution

compare

3.2 Mittag-Leffler's Theorem. *Let G be an open set, $\{a_k\}$ a sequence of distinct points in G without a limit point in G, and let $\{S_k(z)\}$ be the sequence of rational functions given by equation (3.1). Then there is a meromorphic function f on G whose poles are exactly the points $\{a_k\}$ and such that the singular part of f at a_k is $S_k(z)$.*

Proof when $G = \mathbb{C}_\infty$: Since $\mathbb{C}_\infty$ is compact, the set P must be finite and we are done. //

Note : For any compact Riemann surface of genus ≥ 1 (ie. not isomorphic to $\mathbb{C}_\infty$) there exist **Additive** Cousin **Problem** which have no solution

Note : More generally, on any non-compact Riemann surface X, every **Additive** Cousin **Problem** has a solution

- X non-compact Riemann surface $\Rightarrow$ X is a Stein manifold (theorem of Behnke-Stein)
- X Stein manifold $\Rightarrow$ $H^1(X, \mathcal{O}) = 0$ (Cartan Theorem B)
- $H^0(X, \mathcal{O}) \longrightarrow H^0(X, \mathcal{M}) \longrightarrow H^0(X, \mathcal{M}/\mathcal{O}) \longrightarrow H^1(X, \mathcal{O})$ exact

  so $H^1(X, \mathcal{O}) = 0 \Rightarrow H^0(X, \mathcal{M}) \longrightarrow H^0(X, \mathcal{M}/\mathcal{O})$ surjective

  ie. every **Additive** Cousin **Problem** has a solution

Mittag-Leffler's Theorem for $G = \mathbb{C}$: (more precise form)

Let $P = \{a_k\}$ be a sequence (fin. or infin.) of distinct points with no limit point in $\mathbb{C}$ (pole set)

For each k, let $f_k \in \mathbb{C}[z]$ be a polynomial with no constant term

Then for each k, there exists a polynomial $p_k \in \mathbb{C}[z]$ such that

$$f(z) := \sum_{k=0}^{\infty} \left(f_k\left(\frac{1}{z-a_k}\right) - p_k(z) \right) \quad \text{converges locally normally on } G - \{a_k\} .$$

Thus f is a **holomorphic function** on $G - \{a_k\}$

with possible poles at each a_k with principal part $f_k\left(\frac{1}{z-a_k}\right)$

so f is a solution to the **Additive Cousin Problem** for $G = \mathbb{C}$

Proof • Without loss of generality, assume $0 \notin \{a_k\}$ (add $f_0\left(\frac{1}{z}\right)$ later, if needed)

and assume $P = \{a_k\}$ is countable infinite, necessarily with $\lim_{k\to\infty} a_k = \infty$

• For each k, the rational function $f_k\left(\frac{1}{z-a_k}\right)$ is **holomorphic** on $\mathbb{C}$ except for a possible pole at $a_k \neq 0$

so its power series expansion at 0

$$f_k\left(\frac{1}{z-a_k}\right) = \sum_{j=0}^{\infty} c_{a_k,j}\, z^j \quad \text{converges locally normally on } B(0;|a_k|)$$

hence normally on $K_k := \bar{B}\left(0;\frac{|a_k|}{2}\right)$

Thus we can choose $n_k \in \mathbb{N}$ such that (so seq of partial sums conv. unifly on K_k)

$$\left| f_k\left(\frac{1}{z-a_k}\right) - \underbrace{\sum_{j=0}^{n_k} c_{a_k,j}\, z^j}_{=: p_k(z)} \right| \text{ is } \leq \frac{1}{2^k} \quad \text{for all } z \in K_k$$

$p_k(z)$ polynomial function, **holomorphic** on $\mathbb{C}$

• For any compact subset $K \subset G$, there exists $N \in \mathbb{N}$ such that

for all $k \geq N$, one has $K \subset K_k$ (because $\lim_{k\to\infty} a_k = \infty$)

Then $\sum_{k\geq N} \sup_{z\in K} \left| f_k\left(\frac{1}{z-a_k}\right) - p_k(z) \right|$ is $\leq \sum_{k\geq N} \frac{1}{2^k}$ which is $< \infty$

so $\sum_{k\geq N} \left(f_k\left(\frac{1}{z-a_k}\right) - p_k(z) \right)$ converges normally on K (note: all poles a_k with $k \geq N$ are outside K)

• Since $K \subset G$ is arbitrary compact, this shows that

$$f(z) := \sum_{k=0}^{\infty} \left(f_k\left(\frac{1}{z-a_k}\right) - p_k(z) \right) \quad \text{converges locally normally on } G - \{a_k\}$$

#

Digression : Partial fraction expansion of $\pi \cot \pi z$

- The meromorphic function on $\mathbb{C}$ given by

$$f(z) \;:=\; \pi \cot \pi z \;=\; \pi\, i\, \frac{e^{\pi i z} + e^{-\pi i z}}{e^{\pi i z} - e^{-\pi i z}}$$

has the following properties:

(a) f is holomorphic on $\mathbb{C} \smallsetminus \mathbb{Z}$;

(b) f has a simple pole at each $n \in \mathbb{Z}$ with residue 1 ;

(c) f is an odd function of z ;

(d) f satisfies the duplication formula $\boxed{2\,f(2z) = f(z) + f(z+\tfrac{1}{2})}$ for all $z \in \mathbb{C} \smallsetminus \tfrac{1}{2}\mathbb{Z}$.

Pf : (a), (c) are clear ; (b) follows from

$$\lim_{z \to n} \pi\, i\, \frac{e^{\pi i z} + e^{-\pi i z}}{e^{\pi i z} - e^{-\pi i z}} \cdot (z-n) \;=\; 2\pi i \lim_{z \to n} \frac{z-n}{e^{2\pi i z} - 1} \;=\; 1$$

(d) is seen from expanding the right-hand side as

$$\pi \left[\frac{\cos \pi z}{\sin \pi z} + \frac{\cos(\pi z + \frac{\pi}{2})}{\sin(\pi z + \frac{\pi}{2})} \right] \;=\; \pi \left[\frac{\cos \pi z}{\sin \pi z} - \frac{\sin \pi z}{\cos \pi z} \right]$$

$$= \; 2\pi\, \frac{\cos^2 \pi z - \sin^2 \pi z}{2 \sin \pi z \cos \pi z} \;=\; 2\pi\, \cot(2\pi z).$$

#

Goal :

- **Proposition** *One has the partial fraction expansion*

$$\pi \cot \pi z \;=\; \frac{1}{z} + \sum_{n \in \mathbb{Z}_{\neq 0}} \left[\frac{1}{z-n} + \frac{1}{n} \right] \qquad \text{local normal convergence on } \mathbb{C} \smallsetminus \mathbb{Z}$$

$$= \; \frac{1}{z} + \sum_{n \in \mathbb{Z}_{>0}} \left[\frac{1}{z-n} + \frac{1}{z+n} \right] \;=\; \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$$

⚠ Note that it would not make sense to write

$$\pi \cot \pi z \quad \text{as} \quad \sum_{n \in \mathbb{Z}} \frac{1}{z-n}$$

since this series fails to converge absolutely for any $z \in \mathbb{C} \smallsetminus \mathbb{Z}$.

- **Corollary** *One has the partial fraction expansion* (diff. term by term on $\pi \cot \pi z$)

$$\frac{\pi^2}{\sin^2 \pi z} \;=\; \sum_{n \in \mathbb{Z}} \frac{1}{z^2 - n^2}$$

• **Lemma** *The infinite series*

$$g(z) := \frac{1}{z} + \sum_{n\in\mathbb{Z}_{\neq 0}} \left[\frac{1}{z-n} + \frac{1}{n} \right]$$

converges locally normally in $\mathbb{C}\smallsetminus\mathbb{Z}$.

Furthermore,

(a) g *is holomorphic on* $\mathbb{C}\smallsetminus\mathbb{Z}$;

(b) g *has a simple pole at each* $n\in\mathbb{Z}$ *with residue* 1 ;

(c) g *is an odd function of* z ;

(d) g *satisfies the duplication formula* $2\,g(2z) = g(z) + g(z+\frac{1}{2})$ *for all* $z\in\mathbb{C}\smallsetminus\frac{1}{2}\mathbb{Z}$.

Proof. Let $K\subseteq\mathbb{C}\smallsetminus\mathbb{Z}$ be a compact subset and fix $r\in\mathbb{R}_{>0}$ such that $K\subseteq\overline{B(0,r)}$

Then for any $n\in\mathbb{Z}$ with $|n|\geqslant r+1$,

$$\sup_{z\in K}\left|\frac{1}{z-n}+\frac{1}{n}\right| = \sup_{z\in K}\left|\frac{z}{(z-n)\,n}\right| \leqslant \frac{r}{\big||n|-r\big|\,|n|} < \frac{r}{(\,|n|-r\,)^2}$$

and hence

$$\sum_{\substack{n\in\mathbb{Z}\\ |n|\geqslant r+1}} \sup_{z\in K}\left|\frac{1}{z-n}+\frac{1}{n}\right| \quad\text{is}\quad < \sum_{\substack{n\in\mathbb{Z}\\ |n|\geqslant r+1}} \frac{r}{(\,|n|-r\,)^2}$$

$$\text{which is} \quad \leqslant 2r\sum_{k=1}^{\infty}\frac{1}{k^2} < \infty,$$

so $\displaystyle\sum_{n\in\mathbb{Z}_{\neq 0}}\left[\frac{1}{z-n}+\frac{1}{n}\right]$ converges normally on any compact $K\subseteq\mathbb{C}\smallsetminus\mathbb{Z}$ hence locally normally on $\mathbb{C}\smallsetminus\mathbb{Z}$.

Thus,

$$g(z) = \frac{1}{z} + \sum_{n\in\mathbb{Z}_{>0}}\left[\frac{1}{z-n}+\frac{1}{z+n}\right] = \frac{1}{z} + \sum_{n=1}^{\infty}\frac{2z}{z^2-n^2}$$ as well.

Then (a), (b), (c) are clear; for (d), fix $z\in\mathbb{C}\smallsetminus\frac{1}{2}\mathbb{Z}$ and write

$$g(z) + g(z+\tfrac{1}{2}) = \lim_{N\to\infty}\left(\frac{1}{z} + \sum_{n=1}^{N}\left[\frac{1}{z-n}+\frac{1}{z+n}\right] + \frac{1}{z+\frac{1}{2}} + \sum_{n=1}^{N}\left[\frac{1}{z+\frac{1}{2}-n}+\frac{1}{z+\frac{1}{2}+n}\right]\right)$$

$$= \lim_{N\to\infty} \left\{ \frac{2}{2z} + \sum_{n=1}^{N} \left[\frac{2}{2z-2n} + \frac{2}{2z+2n} \right] \right.$$
$$\left. + \frac{2}{2z+1} + \sum_{n=1}^{N} \left[\frac{2}{2z-(2n-1)} + \frac{2}{2z+(2n+1)} \right] \right\}$$

$$= \lim_{N\to\infty} \left\{ 2 \cdot \left(\frac{1}{2z} + \sum_{k=1}^{2N} \left[\frac{1}{2z-k} + \frac{1}{2z+k} \right] \right) + \frac{1}{2z+2N+1} \right\}$$

$$= 2\, g(2z). \qquad \#$$

- **Proposition.** Any *meromorphic function* g *on* $\mathbb{C}$ satisfying (a), (b), (c), (d) is equal to $\pi \cot \pi z$

Proof. Consider $h(z) := g(z) - \pi \cot \pi z$. Then :

(a) h is holomorphic on $\mathbb{C} \setminus \mathbb{Z}$;
(b) h has removable singularities at each $n \in \mathbb{Z}$; } $\Rightarrow$ h is an entire function
(c) h is an odd function of z ; $\Rightarrow$ $h(0) = 0$
(d) h satisfies the duplication formula $2\,h(2z) = h(z) + h(z+\frac{1}{2})$ for all $z \in \mathbb{C}$.

Now apply :

- **Herglotz' lemma** *Let* h *be an entire function satisfying the duplication formula* $2\,h(2z) = h(z) + h(z+\frac{1}{2})$
Then h *is constant of value* $h(0)$

If h is not constant , then by the maximum modulus principle,
there exists $a \in C(0,1) = \partial\overline{\mathbb{D}}$ such that

$$|\,h(z)\,| < |\,h(a)\,| \quad \text{for all } z \in \mathbb{D}.$$

The duplication formula applied to $z = \frac{a}{2}$ yields

$$2\,h(a) = h(\tfrac{a}{2}) + h(\tfrac{a}{2}+\tfrac{1}{2})$$

whence $2\,|\,h(a)\,| \leqslant |\,h(\tfrac{a}{2})\,| + |\,h(\tfrac{a}{2}+\tfrac{1}{2})\,|$

and since $\frac{a}{2} \in \mathbb{D}$ and $\frac{a}{2}+\frac{1}{2} \in \overline{\mathbb{D}}$, this implies

$$2\,|\,h(a)\,| < |\,h(a)\,| + |\,h(a)\,|, \quad \text{contradiction.} \qquad \#$$